To store data and to run applications we use server

Server ::

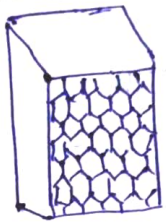
→ server is a physical device which serves the client request
→ server is nothing but a high end version of laptop
→ Laptop / desktop os is different than server operating system
→ server without the operating system called as bare metal server
→ server is used to connect and communicate with external storage devices [JBOD, Storage array]

they are different types of servers

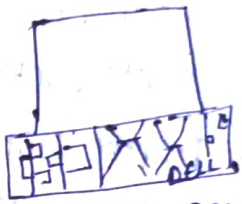
① Rack server
② Blade server
③ Tower server



Rack server



Tower server



Blade server

Form factor of server ::

→ Form factor / foot print
→ we measure server / JBOD / storage Array from factor in UNITS.
→ 1U = = 1.4 inches

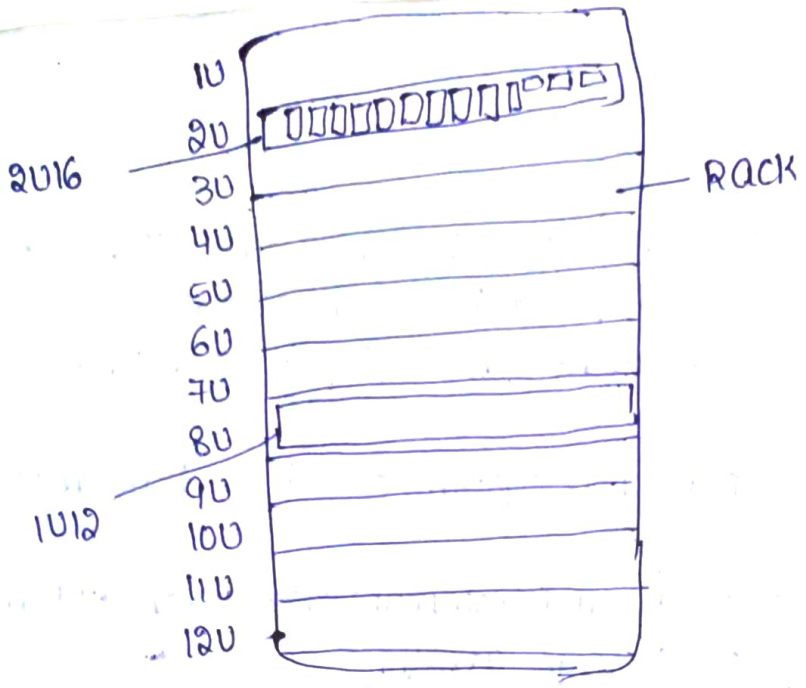
Ex: 2U24 :: 2 units and 24 drive slots
5U84 :: 5 units and 84 drive slots
4U114 :: 4 units and 114 drives

→ Rack is a place which is used to hold storage enclosures / JBOD's / EBODS and servers /storage
→ Racks will be helpful to array the servers in lab enclosures properly
→ It also provides centralized power and network

JBOD :: (Just a Bunch of disks)

→ It is a storage configuration where multiple drives are housed in a single enclosure but treated as independent, seperate storage volumes by os.

Storage Array :: collection of multiple hard drives (HDD's / SSDs) arranged in one system → which provide large amount of data storage and allow multiple servers/computers to access that storage



Physical server components :

Basic components :

① CPU : Multi socket CPU

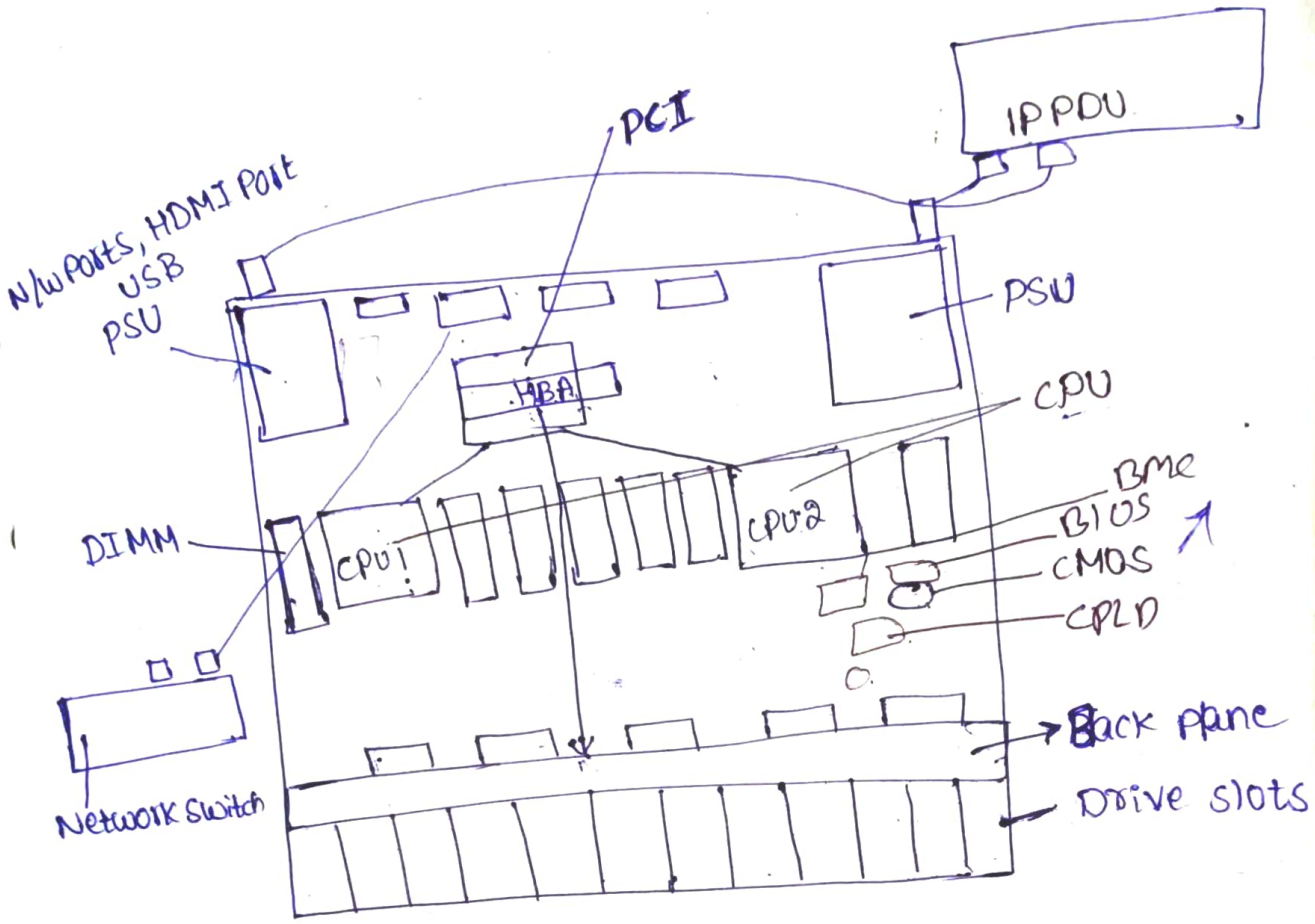
② RAM : DIMM slots

③ Storage : Drive slots

④ PSU : Power supply unit

⑤ Ports : HDMI port, USB ports, Network ports

CMOS → It is small battery on motherboard that keeps BIOS settings and system date/time even when server is power off.



CPU:: (central processing unit)

→ Also called the processor.

→ Its the brain of the computer - executes instructions, performs calculations and control operations.

→ It contains

ALU (Arithmetic Logic unit) - does calculations

CU (control unit) - direct data flow

Registers - small, very fast storage inside CPU.

Cache Memory ::

→ small, high speed memory inside/near CPU to reduce delay in fetching frequently used data.

L1 cache (Level 1):
→ smallest (usually 32KB - 128KB per core)
→ fastest
→ Built directly into the CPU core - inside CPU core

L2 cache (Level 2):
→ Larger (256KB - 2MB per core)
→ slower than L1 but still very fast
→ either private to each core or shared - still very close to CPU

L3 cache (Level 3):
→ Biggest (4MB - 50MB, shared across all cores)
→ slower than L1/L2 but faster than RAM
→ Improves multi-core communication

order of speed : L1 > L2 > L3 > RAM > SSD > HDD
order of size : L3 > L2 > L1

RAM :: (Random Access Memory)

→ Temporary memory where programs/data are loaded while CPU is working
→ volatile - data is lost when power off.

RAM Vs ROM ::

RAM

→ Temporary Memory
→ Data is erased when power is off
→ stores data only while the computer is ON
→ used for running programs and multitasking
→ Ex:: opening chrome, MS word
→ memory

ROM

→ Permanent Memory
→ Data is not erased when power is off
→ stores firmware/software needed to start the computer
→ usually can only read, not written
→ Ex:: BIOS in computer, firmware in washing machine
→ Storage

Types of RAM::

DRAM (Dynamic RAM):

→ Needs to be refreshed thousands of times per second
→ Slower, cheaper, used in main memory

SRAM (Static RAM):

→ Faster, no refresh needed
→ Expensive, used in cache memory

Storage Devices ::

HDD (Hard disk drive)

→ HDD data stores in platters
→ Data stores in form of sectors / blocks
→ Form factor of HDD is 3.5 inches
→ It is less expensive and slower than SSD
→ lifespan of HDD is more than SSD

SSD (Solid State Drive)

→ SSD data stores in flash memory [NAND]
→ Data stores in form of cells / pages / blocks
→ form factor of SSD is 2.5 inches
→ It is more costly and faster than HDD
→ lifespan of SDD is less

Different Types of RAM:: [DDR: (Double data Rate)]

DDR3

→ Introduced in the year 2007
→ Its speed is 800-2133 MT/s (Mega transfer/second)
→ Its voltage is 1.5V
→ power efficiency is moderate
→ usage : Old PCs, laptops, servers

DDR4

→ Introduced in the year 2014
→ Its speed is 2133-3200 MT/s
→ Its voltage is 1.2V
→ power efficiency is better than DDR3
→ usage : Modern PCs, laptops, servers

DDR5

→ Introduced in the year 2020
→ Its speed is 4800-8400+ MT/s
→ Its voltage is 1.1V
→ power efficiency is best
→ usage : Latest gaming PCs, high-end servers, new laptops

Microcontroller Vs Microprocessor::

Microprocessor	Microcontroller
→ Just the CPU (needs external RAM, ROM, I/O)	→ CPU + RAM + ROM + I/O ports on a single chip
→ Ex:: Intel i7, AMD Ryzen	→ Ex:: Arduino, PIC, ARM cortex-M
→ used in PCs, servers, laptops	→ used in embedded system (washing machines, cars, IOT devices)

CPU core:: → physical hardware (real processing unit inside CPU)

→ A core = a mini CPU inside the processor
→ Each core can run its own task [fetch, decode, execute instructions]
→ Ex: Quad-core CPU = 4 cores (can handle 4 tasks at once)
→ In modern CPUs a chip contain multiple cores so it is called "multi core processor"

CPU Thread::

→ A thread is a virtual version of a core. It represents the tasks that the core can handle. CPU uses threads to perform multiple operations at once
→ Many CPUs use Hyper-Threading / SMT (Simultaneous Multi-threading):
  → 1 core → 2 threads
  → Ex: 4-core, 8-thread CPU can handle 8 tasks at once

Multi Socket CPU::

→ A server motherboard can have more than one CPU socket.
→ Ex: A dual socket server has 2 physical CPUs, each with multiple cores
→ used in high-performance servers for big workloads (database, AI, cloud)

GPU (Graphics processing unit)::

→ specialized processor designed for parallel processing.
→ originally for graphics but now used in AI, deep learning, 3D rendering, scientific computing
→ Has thousands of smaller cores optimized for parallel tasks

CPU Vs GPU:

Feature	CPU	GPU
Cores	- Few (4-64 powerful cores)	- Thousands of small cores
Best for	- Sequential tasks, logic, OS, apps	- parallel tasks, graphics, AI, ML
Example	- Running windows, browsing, databases	- Gaming, training AI models, mining crypto

PSU (Power supply unit)::

→ The PSU is like the heart of a computer. It supplies power to all components
→ converts AC (Alternating current) from wall socket into DC (Direct current) needed by computer parts
→ supply stable power to CPU, GPU, motherboard, storage etc

AC Vs DC::

AC (Alternating current):
→ Direction of current keeps changing
→ comes from power outlets (home electricity)

DC (Direct current):
→ flows in one direction
→ used inside electronic devices (CPU, GPU, RAM)

X:86 - 8086 Microprocessor::

8086 Microprocessor:
→ Made by Intel in 1978
→ It is 16-bit processor (can handle 16-bits of data at a time)
→ Has 20-bit address bus → can access 1MB of memory
→ used in early IBM computers
→ It is the father of modern Intel processors

X86:
→ the family of processors that started with 8086
→ the "86" comes from the last two digits (8086, 80286, 80386, 80486)

→ Todays Intel and AMD processor (like core i7, Ryzen) are still based on x86 architecture
→ supports CISC (complex instruction set computing) → many powerful instructions

From the diagram ::

Drive slots ::
→ Drive slots are used to hold the drives in server

Backplane ::
→ Backplane is a device that allows drives [HDD, SSD] to connect to the server PCI.
→ Backplane is also called as sideplane (or) midplane based on the server/storage enclosure architecture

PCI ::
→ PCI stands peripheral component Interconnect which allows external devices to communicate with CPU
Ex: HBA [host bus adapter]
Network cards
Graphic cards
NVME devices
GPU
CXL devices

Heat Sink ::
→ Both the CPU's will be covered with heat sinks
→ Heat sinks will be used reduce heat produced by CPU's

HBA ::
→ HBA stands for Host Bus adapter which is used to connect external storage devices to PCI with PCI cards
Ex: HDD, SSD, Backplane, External storage enclosure

PSU ::
→ It is used to supply
→ PSU will take the power from external power sources and supplies to the server components
→ which is also responsible for converting power from AC to DC.
→ In an average server requires 1100 W to 1600 W

→ Servers will have redundance power supply [PSU]

PCI ::

→ PCI will be connected to CPU with I2C bus → cable

→ PCI & CPU are connected with PCI lane

PCI lane ::

→ PCI is an full duplex protocol

→ Each PCI lane is capable of sending and receiving the data at a time.

→ They are different Generations of PCI

Gen1, Gen → speed → 250MB :: 2.5 GT/s [Giga transfer per second]

Gen2 → 500 MB :: 5 GT/s

Gen3 → 1GB :: 8 GT/s

Gen4 → 2GB :: 16 GT/s

Gen5 → 4GB :: 32 GT/s

Gen6 → 8GB :: 64 GT/s

Gen7 → 16GB :: 128 GT/s

Bandwidth ↓ speed

→ the current generation of PCI is PCI Gen7

PCIE and PCIX ::

feature	PCIX	PCIE
fullform	- Extended	- Express
Type	- parallel	- Serial
Max Bandwidth	- ~1 GB/s	- upto 128 GB/s
scalability	- limited	- highly scalable
usage	- legacy servers	- Modern pcs, server storage, GPUs

→ 1 lane = 2 direction communication (sending & receiving)

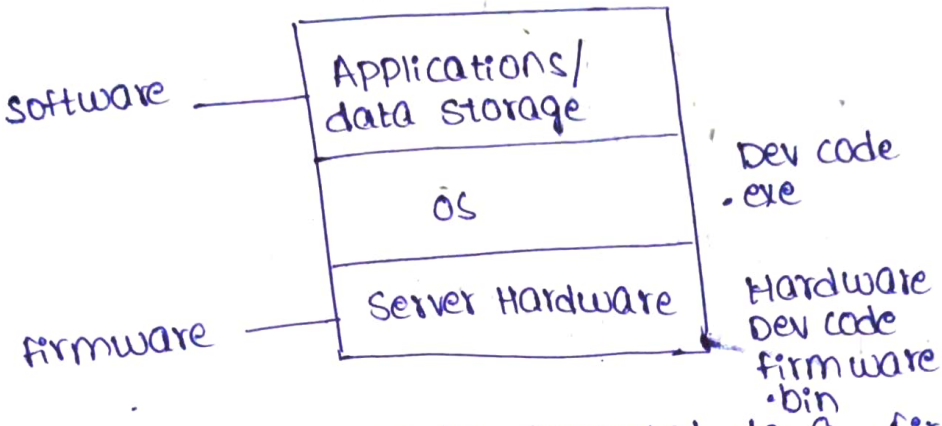
→ If there are many lanes the performance and speed will increase

Firmware :: → Software programmed into hardware.

→ firmware is a software (developer code) which resides inside the hardware devices

→ firmware defines the functionality of a hardware devices

→ without firmware we cannot use any hardware devices



→ Developer code will be converted to a firmware image (.bin files)

BIOS ::

→ BIOS stands for Basic Input/output system

→ BIOS is a firmware which resides inside the server motherboards

→ when we power on the server first control goes to server BIOS will initialize all the hardware components to server

→ BIOS will responsible for operation called POST (Power On Self Test)

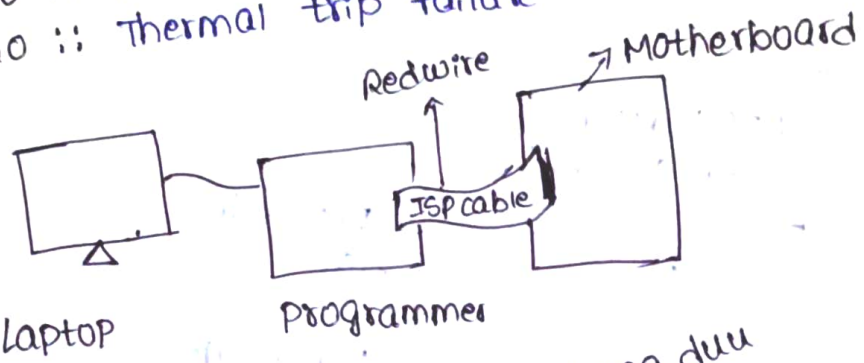
→ BIOS will check all the hardware components in the server are coming up or not

→ If any ~~device~~ components is not coming up it will through the POST code Error (In older desktops beep errors)

POST codes ::

0000 :: Time issue
0003 :: Battery low
000C :: RAM failed
000E :: Drive failed
8601 :: BMC Not responding

8102 :: USB port failed to initialize
005D :: SMART Error
0041 :: Display memory test failed
005D :: SMART
005E :: password
0160 :: processor missing microcode
0120 :: Thermal trip failure



To install BIOS firmware → By using duu

1. OS level
2. Special hardware [Dedi prog]

→ when we update firmware, EEPROM will be programmed
After updating BIOS firmware server will go for hard reboot

Difference between BIOS and UEFI

UEFI ::
→ UEFI stands for unified Extendible firmware Interface

BIOS	UEFI
→ BIOS stands for Basic input and output system	→ UEFI stands for unified extendible firmware Interface
→ BIOS provide slower boot time	→ UEFI provides faster Boot time
→ BIOS uses MBR partition table (Master Boot Record)	→ UEFI used GPT Partition table (Guid partition table)
→ BIOS supports drive size upto 2.2 TB	→ UEFI supports drive size upto 9 ZB
→ system with BIOS are less secure as it dont have secure boot feature	→ systems with UEFI are more secure as it is having "secure boot" feature

→ BIOS store information about initialization in firmware

→ BIOS will load Boot loader from the boot device

→ UEFI stores information about initialization and starts up in .efi file

other options operations from BIOS::

→ set BOOt order / boot priority
→ set Date & Time
→ Discover hardware
→ Virtualization setting
→ Discovering BMC IP
→ converting drives from RAID/ Non RAID or AHCI Mode

BMC::

→ BMC stands for baseboard management controller
→ BMC is a small chip that is embedded in the mother board.
→ BMC is mainly used for remote management of a server/ storage array
→ without login to the server/storage array we can control them by using BMC
→ using BMC we can monitor the health and status of sensors and all the components of a server/storage array
→ If CPU gets crashed also, we can login to the system via BMC.

1. Remote Management
2. Monitor the servers
   Health status of servers
3. It will capture all the events
4. List all Hardwares [FRU] & sensors
5. Power operations
6. Access to baremetal servers
7. Install OS
8. Install firmwares
9. CLI & GUI
10. support for different interface
    IPMI, Redfish, Swordfish, Postman etc...

Server OEMS BMC Names ::

Dell : Idrac (Integrated Dell Remote Access controller)
HP : ILO (Integrated Lights-out)
CISCO : CIMC (Cisco Integrated Management controller)
Lenovo : Xclarity controller
Super Micro : Super BMC
Intel : RMM / BMC
Inspur : iBMC
IBM : IMM

How remote management is possible
BMC have multiple interfaces to control the system
1. IPMI [Intelligent platform management Interface]
→ (a) IPMI is a protocol that enabled the communication between CPU and BMC
→ (b) It provides management and monitoring capabilities independently of the CPU
→ (c) IPMI interface uses IPMB protocol
→ (d) the current version of IPMI is IPMI V2.0
→ (e) there are 2 types of IPMI interface are available
   (i) inband
   (ii) out of band.

Inband :: / Interacting with BMC directly

→ If we execute the commands, ~~directly on BMC it is~~ from target is called as In band

out of band ::

→ If we execute the commands/ ~~via~~ LAN Interacting with BMC via LAN is called out of band

Generic Examples:

① Inband :: SSh, RDP, telnet etc

② out of band :: KVM, IPMI, Redfish etc...

You can do the ~~follow~~ in.

fru → field Replacable unit

IPMI tools Inband commands ::

ipmitool chassis status :: To check enclosure/ server status

ipmitool chassis power [on][off|cycle] :: for power related operations

ipmitool fru print :: To print all components in a server

ipmitool sensor list :: It lists all the sensor related info

ipmitool sdr list :: To get logs about sensors

↳ sensor data repository

ipmitool sel list :: To get all the system event logs

ipmitool mc info :: Show BMC info

ipmitool mc reset :: " " reset

ipmitool bmc reset cold :: Reset BMC

ipmitool bmc reset warm :: " "

ipmitool lan print 1 :: Show LAN settings

ipmitool lan set 1 ipaddr 192.168.1.100 :: Set LAN IP address (for BMC)

ipmitool lan set 1 netmask 255.255.255.0 :: " " " " " "

ipmitool lan set 1 defgw ipaddr 192.168.1.1 :: " " " " " "

ipmitool lan set 1 access on :: Enable LAN access

IPMI tools outband command ::

ipmitool -H <IPaddress> -I lanplus -u <username> - P <password> mc info

ipmitool -I <lanplus> -H <ip_address> -u <username> -p <password> mc info

↳ BMC-IP

→ mc reset

→ chassis status :: To check enclosure

→ chassis power [on|off|cycle] ::

→ fru Print ::

→ Sensor list ::

→ sdr list
→ sel list
→ bmc reset cold :: Reset BMC
→ bmc reset warm:: " "
→ lan print 1 :: Show LAN settings

→ IPMI tool works based on serial over LAN

Redfish::
→ Redfish is an RESTful API designed for managing servers Datacentre products
→ Redfish works based on Http protocol

Redfish follows HTTP Methods
1. GET :: Retrieve data from the server. It will get original data
2. POST :: Create a new resource on the server
3. PUT :: Update an existing resource or create a new resource if it doesn't exist
4. PATCH :: Update an existing resource
5. DELETE :: Remove a resource from the server

HTTP Response codes ::
400 → Client side failure
500 → Server side failure
100, 200 → Success codes

→ In Redfish data will be exchanged in JSON format
→ It is one of the interface for accessing BMC
→ It can be used with POSTMAN, Talend API & CURL

use cases ::
① server Management ::
→ Monitor hardware (e.g., temperature, fan speed)
→ Power on/off/reboot servers
→ Manage BIOS / UEFI settings

→ configure network interfaces
→ Manage storage (drives, volumes)
→ update firmware
→ Access logs/events

we need enable Redfish from BMC console

Redfish syntax :: /redfish/v1/systems

systemID :: 1

GET /redfish/v1/systems :: To get system ID

GET /redfish/v1/systems/{systemId} :: To get information about particular system

POST /redfish/v1/systems/{systemId}/Actions/computersystem.Reset

```
{
  "ResetType":"on"
}
```

CURL ::

→ command to get responce from http based servers

→ we need to execute curl command from windows or Linux Terminals

To get Redfish version ::

curl -u username:password -K https://<BMC_IP>/redfish/v1/

To list all systems ::

curl -u username:password -K https://<BMC_IP>/redfish/v1/systems

To get details of specific system ::

curl -u username:password -K https://<BMC_IP>/redfish/v1/systems/{systemId}

Power operations ::

curl -u username:password -K -X POST https://<iDRAC_IP>/redfish/v1/Systems/{systemId}/Actions/computersystem.Reset\ -d '{"ResetTPye":"on"}'

To get power status ::

curl -u username:password -K https://<iDRAC_IP>/redfish/v1/systems/{systemId}/power

To list BIOS settings ::

curl -u username:password -K https://<iDRAC_IP>/redfish/v1/systems/{systemId}/BIOS

Update BIOS settings ::

```
curl -u username:password -k -X PATCH https://<iDRAC_IP>/
     redfish/v1/systems/{systemId}/BIOS \-d
     '{"Attributes":{"somethin setting":"NewValue"}}'
```

- To list all physical drives ::

```
curl -u username:password -k https://<iDRAC_IP>/
          redfish/v1/systems/{systemId}/storage
```

CPLD ::

→ CPLD stands for complex programmable logic devices, which is a small chip embedded in the server mother-board.

→ CPLD is used to supply power to all the sensors in the motherboard and It is also responsible for controlling the power fluctuations in the server

→ CPLD has its own firmware, so when we update CPLD firmware server goes for hard reboot.

component	Effect on the server
→ drives	- No effect
→ Backplane	- No effect
→ HBA	- No effect
→ Right controller	- No effect
→ BMC	- ~~No effect~~ soft reboot
→ BIOS	- Hard reboot
→ CPLD	- hard reboot
→ PSU	- No effect
→ Fans	- No effect
→ Network card	- No effect

Driver ::

→ Software that helps as talk to hardware

→ Installed in os

→ Acts as translator

1xx → Info
2xx → Success
3xx → Redirect
4xx → client error
5xx → server error

```
l1 = [a, b, c, d, e, f, g]
     0  1  2  3  4  5
l1[:4]
```

200 (Success codes

200 - Success
201 - Resource created
202 - Processing later
204 - Success (No body, No content)

203

300 (Redirect)

301 - Moved permanently
302 - found (temperory redirect)
304 - Not modified

303 300

400 (Client error)

400 - Bad request (Invalid request)
401 - unauthorized (Not authorited)
403 - forbidden (NO access)
404 - Not found (Resource missing)

402

500 (server error)

500 - Server failure
501 - feature not supported
502 - upstream error
503 - Server down
504 - upstream timeout
~~505~~ -

GET → Get the details
POST → create new resource
PATCH → update existing resource
PUT → full replace
DELETE → Remove